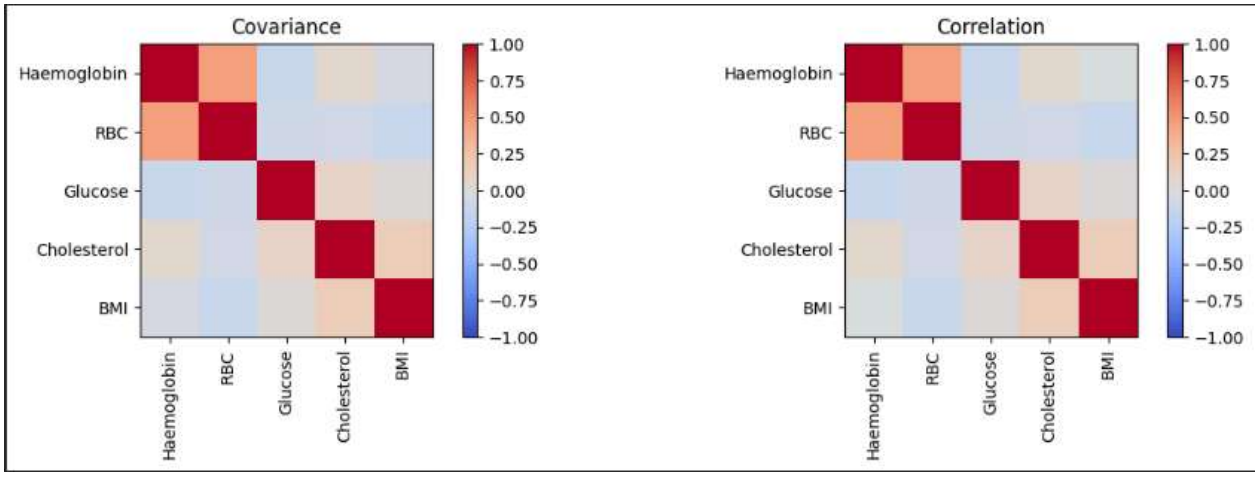
EXPERIMENT2: Covariance, Eigenvalues and Eigenvectors Analysis
Problem Statement: A hospital lab records several correlated blood parameters per patient. Analysts
must know which parameters move together and which directions carry most of the variation. A dataset
of 200 patient records with haemoglobin, RBC count, glucose, cholesterol and BMI. The lab suspects
redundancy among these readings. The objective is to compute covariance and correlation, then extract
eigenvalues and eigenvectors to find the principal directions of variation. Use NumPy, Pandas, Matplotlib.
Tasks:
1. Load the data into Pandas and standardise each feature.
2. Compute the covariance and correlation matrices; plot them as heatmaps.
3. Compute eigenvalues and eigenvectors with NumPy; sort by eigenvalue.
4. Plot the data in 2-D with the top two eigenvectors drawn as arrows.
5. Report the percentage of variance captured by each eigenvalue.

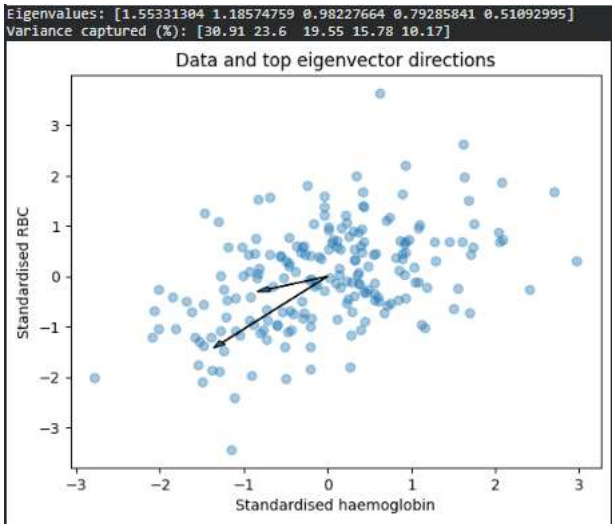
AIM:
To calculate covariance and correlation matrices for patient data and identify the principal directions of variation using eigenvalues and eigenvectors.
PROGRAM:

```
df = pd.DataFrame({
 'Haemoglobin':np.random.normal(13,1.5,200),
 'RBC':np.random.normal(4.8,.5,200),
 'Glucose':np.random.normal(100,15,200),
 'Cholesterol':np.random.normal(190,25,200),
 'BMI':np.random.normal(25,4,200)})
df['RBC'] += (df['Haemoglobin']-13)*.15
z = (df-df.mean())/df.std(ddof=0)
cov, corr = z.cov(), z.corr()
fig,ax=plt.subplots(1,2,figsize=(12,4))
for a,m,t in zip(ax,[cov,corr],['Covariance','Correlation']):
 im=a.imshow(m,cmap='coolwarm',vmin=-1,vmax=1);
a.set_xticks(range(5),m.columns,rotation=90);
a.set_yticks(range(5),m.columns); a.set_title(t); fig.colorbar(im,ax=a)
plt.tight_layout(); plt.show()
vals, vecs=np.linalg.eigh(cov); order=np.argsort(vals)[::-1];
vals,vecs=vals[order],vecs[:,order]
print("Eigenvalues:", vals)
print("Variance captured (%):", np.round(vals/vals.sum()*100,2))
```

```
plt.scatter(z.iloc[:,0],z.iloc[:,1],alpha=.4)
center=z.iloc[:,:2].mean().values
for k in range(2):
 plt.arrow(center[0],center[1],vecs[0,k]*2,vecs[1,k]*2,head_width=.1)
plt.xlabel('Standardised haemoglobin'); plt.ylabel('Standardised RBC');
plt.title('Data and top eigenvector directions'); plt.show()
```

OUTPUT:





RESULT:

Covariance and correlation matrices were successfully calculated and visualised. Eigenvalues, eigenvectors and explained variance percentages were obtained to understand the variation in the dataset.